Mod 5

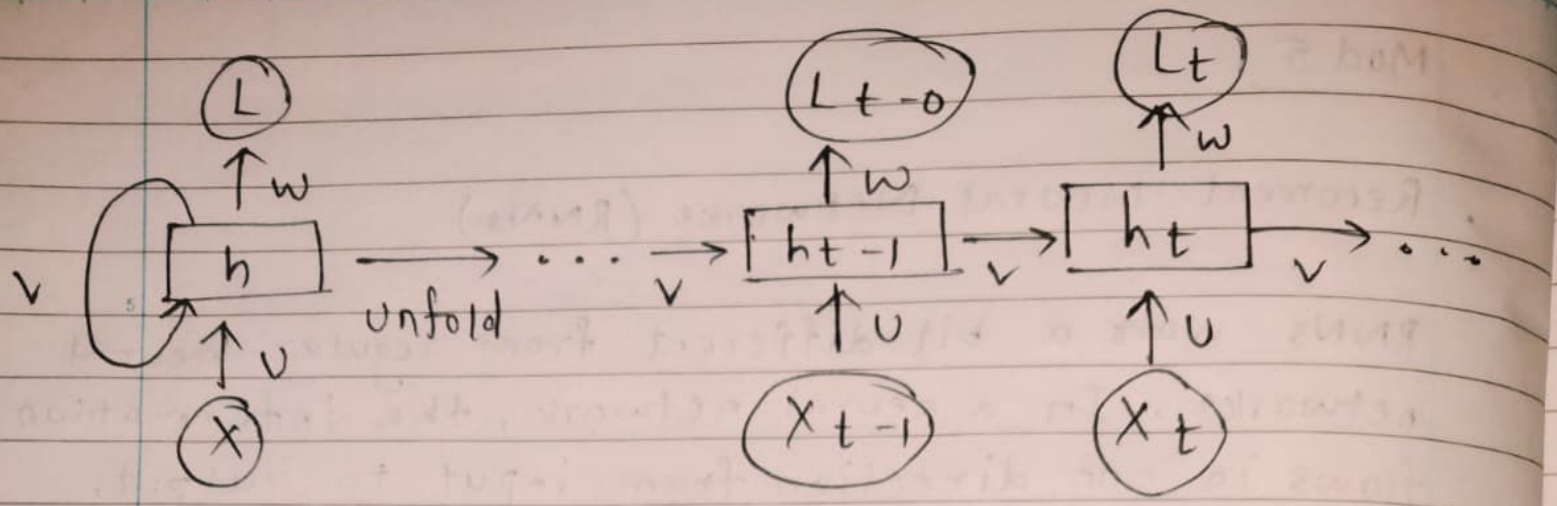
. Recurrent Neural Networks (RNNs)

RNNs work a bit different from regular neural networks. In a neural network, the information flows in one direction from input to output. However, in RNN, information flows ~~in one~~ is fed back into the system after each step.

Think of it like reading a sentence, when you are trying to predict the next word you don't just look at the current word, but also need to ~~rembe~~ remember the words that came before to make an accurate guess.

RNNs allow the network to remember past information by feeding the output from one step into next step. This helps the network understand the context of what has already happened and make better predictions based on that.

For example, when predicting the next word in a sentence, the RNN uses the previous words to help decide what word is most likely to come next.



① Recurrent Neurons

The fundamental processing unit in RNN is a Recurrent Unit. Recurrent Units hold a hidden state that maintains information about previous inputs in a sequence.

② RNN unfolding

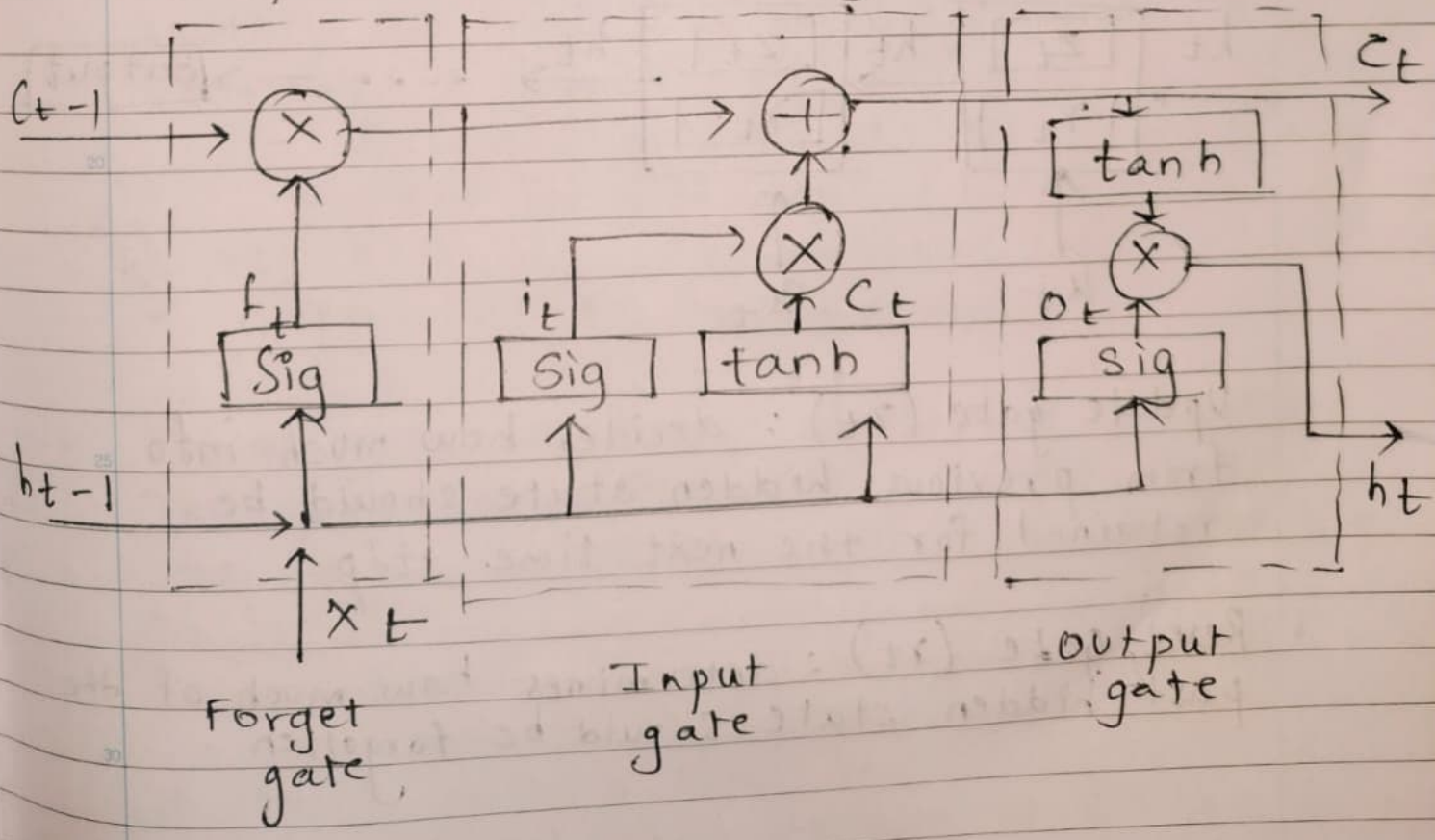
RNN unfolding is the process of expanding the recurrent structure over time steps. During unfolding each step of the sequence is represented as a seperate layer in a series illustrating how information flows across each step.

- Long Short Term Memory (LSTM)

LSTM is an enhanced version of RNN that can capture long-term dependencies in sequential data making them ideal for tasks like language translation, speech recognition and time-series forecasting.

Architecture:

1) Input Gate: controls what information is added to the memory cell.
2) Forget Gate: determines what information is removed from the memory cell.
3) Output Gate: controls what information is output from the memory cell.



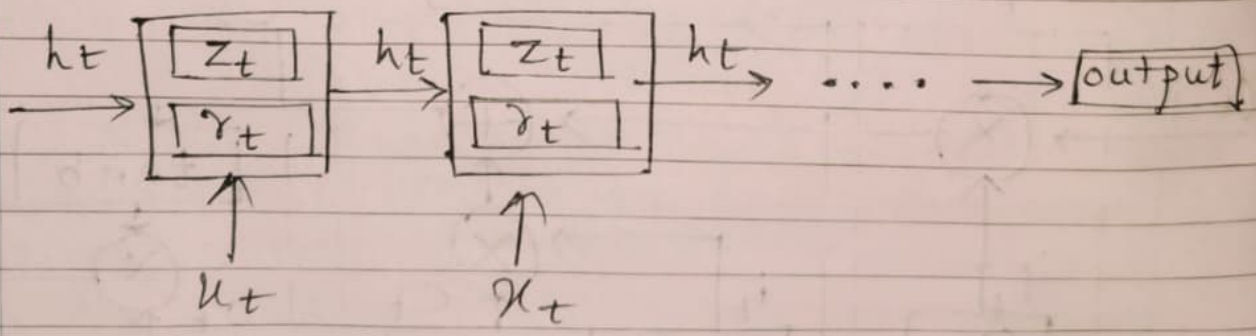
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Applications of LSTM

1. Language Modeling
2. Speech Recognition
3. Time Series Forecasting
4. Anomaly Detection
5. Recommender Systems.

• Gated Recurrent Units

GRUs are a type of RNN that use gating mechanisms to selectively update the hidden state at each time step allowing them to remember important information, while discarding irrelevant details.



Update gate (z_t): decides how much info from previous hidden state should be retained for the next time step.

Reset Gate (r_t): determines how much of the past hidden state should be forgotten.

LSTM	GRU
3 gates	2 gates
cell state	hidden state only
slow training speed	faster
higher complexity	lower complexity
better for long-term memory	better for short-term memory

- vanishing gradients

The gradients (partial derivatives of the loss function wrt weights) become very small as they are backpropagated through the network.

  - Slow or stalled learning
  - Poor performance

× use ReLU
× careful weight initialization

- Exploding gradients

The gradients become very large causing the weights to grow uncontrollably leading to instability.

  - NaN weights
  - Unstable training

× lower learning rates
× use LSTMs / GRUs instead of RNNs

- Problems during RNN training

1. Vanishing gradient
2. Exploding gradient
3. Long Training Time
4. Difficulty in capturing long term dependencies
5. Overfitting

- Ill conditioning

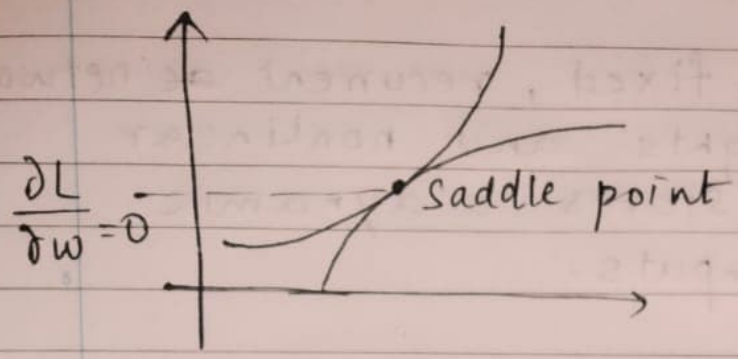
Ill conditioning refers to a situation where a small change in input causes a disproportionately large change in output or solution.

- Reasons
  - ratio of largest and smallest singular value is very high
  - features are on different scales
  - poor learning rates
- Solution
  - normalization
  - dimensionality reduction
  - regularization

- Saddle points

A saddle point is a point in the optimization landscape where the gradient is zero but the point is neither a local minimum nor a local maximum.

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- Bidirectional RNNs

BiRNN is an extension of the standard RNN that can capture both past and future context in a sequence.

There are two RNNs,

Forward RNN that processes sequence from $t = 1$ to T.

Backward RNN that process sequence from $t = T$ to 1.

- Echo State Networks (ESN)

ESNs are a type of RNN designed to overcome some of the training difficulties of traditional RNNs especially vanishing and exploding gradients.

Instead of training all the weights in an RNN, ESNs only train the output layer.

Architecture:

(1) Input layer : connects input to reservoir

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(2) Reserroir - large, fixed, recurrent ~~ac~~ network with random weights and nonlinear activations that stores a dynamic "echo" of past inputs.

(3) Output layer - only this part is trained.

Echo state means the influence of a past input fades over time. The current state should primarily reflect recent inputs. It is ensured by controlling the largest absolute eigenvalue of the reservoir weight matrix

- Meta Algorithms

Meta algorithms are higher-level algorithms designed to combine, modify, or optimize learning algorithms to improve performance, flexibility and robustness.

(1) Ensemble methods
combine multiple models
(2) Hyperparameter Optimization
(3) Cross validation
multiple train-test splits
(4) Model Selection